



Hardware and Systems

Training Guide

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Hardware and Systems

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About this Guide

This document provides insight and overview into the many different hardware and software products we sell.

Who Should Use It

This document is intended for sales and service representatives who need to better understand what a system is and the product make up.

Typographical Conventions

This document uses the following typographical conventions:

Bold – Important aspects

Red Highlighted – Important Note that required additional attention

Italics – is a demonstration or example

Bulleted – Identifies important elements

Numbered – Step-by-Step instructions on procedures

Notes:

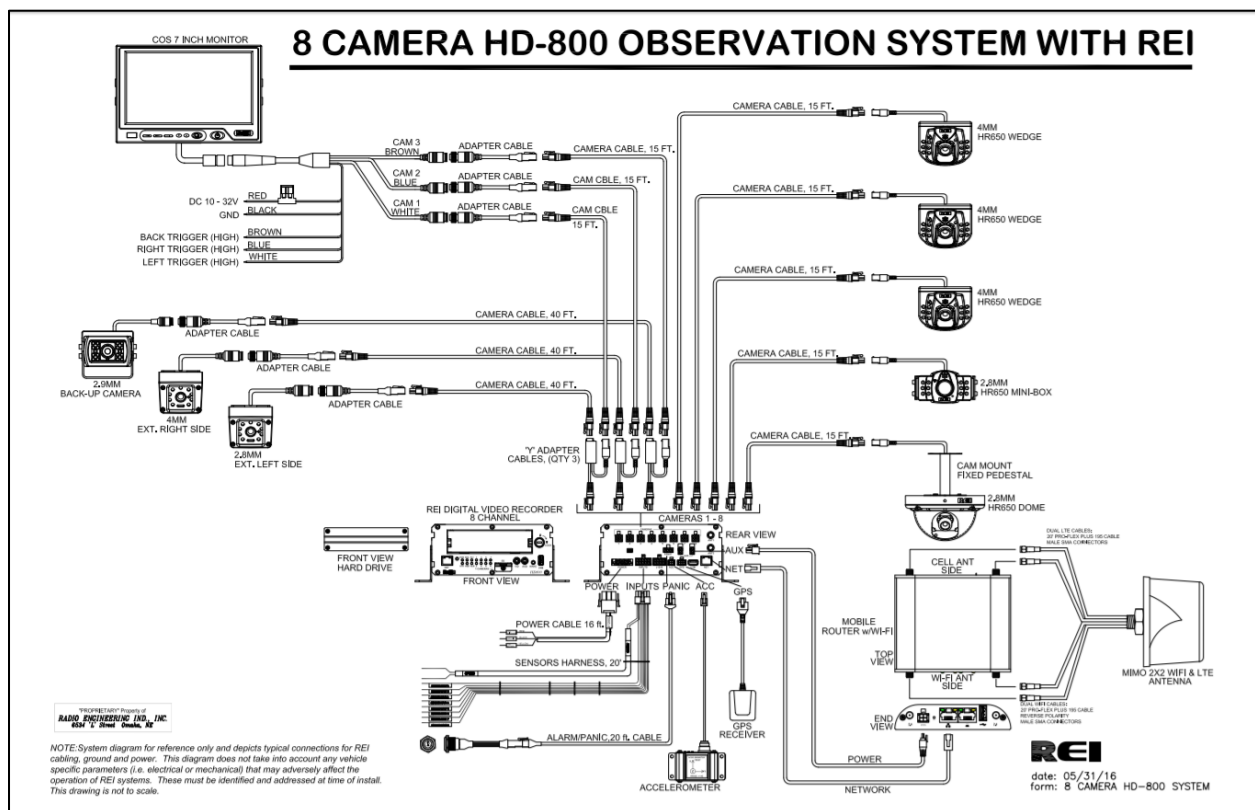
1 System

Let us first take a moment to better understand what a specific “**System**” includes. All basic systems come complete and include:

- Digital Video Recorder (DVR),
- Removable Hard Drive
- High-resolution color cameras,
- Software, and
- All accessories necessary to ensure systems are ready to run upon installation.

Note: Each of the items from a system can be purchased separately as replacements and each system is configurable per the specific needs of the bus. Additionally, there are add-on features that can be purchased separately per the requirements of the school or state laws.

Below is an example of a complete 8-Camera High Definition Observation System.



2 Hardware

2.1 Digital Video Recorder (DVR)

There are currently 5 different classes of DVR offered by REI. (*Our flagship product. Will soon replace the HD800 and 420 systems.)

- *HD5
 - HD5-1200 (12 Channels)
 - Replaces HD800
 - HD5-600 (6 Channels)
 - Replaces HD420
- **HD800 – 8 Channel, Uses Removeable Hard Drive and SD Card**
- **HD420 – 4 Channel - Uses Removeable Hard Drive and SD Card**
- **DR40 – 4 Channel - Uses only a removable solid-state Hard Drive**
- **SD40 – 4 Channel - Uses only an SD Card**



There are numerous variations of each product line. The specific options you select will be based upon the needs of the school district, state laws, as well as the type of bus.

Your standard part number for the DVR systems are broken out as such:

- **HD800-8-2TB** – This HD800 DVR is an 8 Channel DVR / comes with 8 Cameras / and a 2 TB Hard Drive.
- **SD40-4-32GPS** – This SD40 DVR is a 4 Channel DVR / comes with 4 Cameras / and a 32 GB SD Card / and GPS Receiver

Note: You may also see an additional designator behind the model.

For example:

- **ALPR** = Automated License Plate Recognition
- **STCAM** = Stop Arm Cam
- **W** = WiFi
- **SAW** = Stop Arm with WiFi
- Etc.

These are all specific applications that can be added to the basic units.

To select the correct DVR configurations for their school's bus you will need to ask them for specifics on how they intend to use their new system.

For example:

- How many cameras?
- What type of cameras?
- Quality vs. Quantity in terms of pictures. That will help determine hard drive size.
- Wi-Fi Capable, cellular, or just pull and play.
- Will they require internet access?

For example, the HD420-2 comes with 2 cameras, whereas the HD800-6 comes with 6 cameras.

Notes:

2.2 Hard Drives

In this section, the focus turns to the different Hard Drives. There are 3 main types of hard drives available. They include:

- **Removable Hard Drive** - The Removable Hard Drives storage sizes range from 320 Gb up to 2 Tb
- **Solid State Drive** – The Removable Hard Drives storage sizes range from 500 Gb up to 2 Tb
- **Removable SD Card** - The SD Cards storages sizes range from 32 Gb, 64 Gb, up to 256 Gb.



To sell a DVR, you will need to know which hard drive to select for their specific application. The best way to begin is to determine their specific use.

For example: Are they looking for image clarity, recording time, or storage space? To determine the best fit for them use the **Record Time Estimator** found here:

Record Time Estimator HD5 Drives (Application): www.radioeng.info/rte/hd5.zip

Record Time Estimator: <http://www.radioeng.info/rte/>

This will aid in your ability to best select the drive that meets their needs.

For example: When you input all the specifications for a HD800 2Tb Drive it will show a record time of approximately 592 hours. However, if you go with the same system and change the configuration for best quality images, you end up with a record time of approximately 148 hours. Better picture, but less record time. Will this be a factor to your client?

Notes:

3 Cameras

In this next section, the focus will be on the variety of different camera options available. Camera placement and options are a critical component of designing a proper system, and this is one section that will take the most time to learn about. Each camera has a specific function and location based off the lens, style, and features. This will be further simplified in this section.

These are the different options of cameras REI supplies.

- Minibox Camera 2.4 mm Lens 2.8 mm Lens 4.0 mm Lens 6.0 mm Lens 8.0 mm Lens 	- Dome Camera 2.8 mm Lens 4.0 mm Lens 6.0 mm Lens 8.0 mm Lens 
- Exterior Side Mount Cameras 4.0 mm Lens 6.0 mm Lens 8.0 mm Lens 12.0 mm Lens 16.0 mm Lens 	- Rear/Side Door 2.8 mm Lens 
- Recessed Camera 2.8 mm Lens 3.6 mm Lens 	- Stop-Arm Cameras-ALPR 2 Lane 4 Lane 
- Wedge Camera 2.8 mm Lens 4.0 mm Lens 6.0 mm Lens 8.0 mm Lens 	

To determine the best camera for the application you must first understand more about viewing angle and distance. Each camera has an optimal viewing angle and distance. That distance is defined by the type of lens it uses.

For example: a 2.8mm lens would have a viewing angle of 09° horizontal, 82° vertical, and 136° diagonal.

Lens Size	Viewing Angle (Approximately)			Distance to Object / Size of picture (Width X Height)											
	Horizontal °	Vertical °	Diagonal °	5 Ft		10 Ft		15 Ft		25 Ft		50 Ft		100 Ft	
2.8mm	09°	82°	136°	14.0'	10.5'	8.0'	21.0'	42.0'	31.5'	70.0'	52.5'	140'	105'	280'	210'
3.6mm	74°	56°	92°	7.5'	5.7'	15.0'	11.4'	22.5'	17.1'	37.5'	28.5'	75'	57'	150'	114'
4.0mm	67°	50°	83°	6.6'	5.0'	13.2'	10.0'	19.8'	15.0'	33.0'	25.0'	66'	50'	146'	110'
6.0mm	42°	32°	53°	3.8'	2.9'	7.6'	5.8'	11.4'	8.7'	19.0'	14.5'	38'	29'	76'	58'
8.0mm	32°	24°	40°	2.9'	2.2'	5.8'	4.4'	8.7'	6.6'	14.5'	11.0'	29'	22'	58'	44'
12mm	22°	16°	28°	1.9'	1.5'	3.8'	2.8'	5.7'	4.5'	9.5'	7.5'	19'	15'	38'	28'
16mm	16°	12°	21°	1.4'	1.1'	2.8'	2.2'	4.2'	3.3'	7.0'	5.5'	14'	11'	28'	22'

As you can see from the image below you get a very wide view covering a large service area, but not a lot of detail when it comes to the automobile.

2.8mm Example:



Changing the lens size to a 12mm lens would have a viewing angle of 22° horizontal, 16° vertical, and 28° diagonal.

Lens Size	Viewing Angle (Approximately)			Distance to Object / Size of picture (Width X Height)											
	Horizontal °	Vertical °	Diagonal °	5 Ft		10 Ft		15 Ft		25 Ft		50 Ft		100 Ft	
2.8mm	09°	82°	136°	14.0'	10.5'	8.0'	21.0'	42.0'	31.5'	70.0'	52.5'	140'	105'	280'	210'
3.6mm	74°	56°	92°	7.5'	5.7'	15.0'	11.4'	22.5'	17.1'	37.5'	28.5'	75'	57'	150'	114'
4.0mm	67°	50°	83°	6.6'	5.0'	13.2'	10.0'	19.8'	15.0'	33.0'	25.0'	66'	50'	146'	110'
6.0mm	42°	32°	53°	3.8'	2.9'	7.6'	5.8'	11.4'	8.7'	19.0'	14.5'	38'	29'	76'	58'
8.0mm	32°	24°	40°	2.9'	2.2'	5.8'	4.4'	8.7'	6.6'	14.5'	11.0'	29'	22'	58'	44'
12mm	22°	16°	28°	1.9'	1.5'	3.8'	2.8'	5.7'	4.5'	9.5'	7.5'	19'	15'	38'	28'
16mm	16°	12°	21°	1.4'	1.1'	2.8'	2.2'	4.2'	3.3'	7.0'	5.5'	14'	11'	28'	22'

As you can see, the image field of view is greatly reduced. However, the resolution has increased and much more of the car is visible.

12mm Example:



Of course, this is just an example and based off generic numbers. The view from our REI cameras may be slightly different, but the purpose of this is just to understand what you are

looking at. To learn more about specific products and their actual numbers you may review the technical breakdowns on our share drive location: **F:\PDF\7**

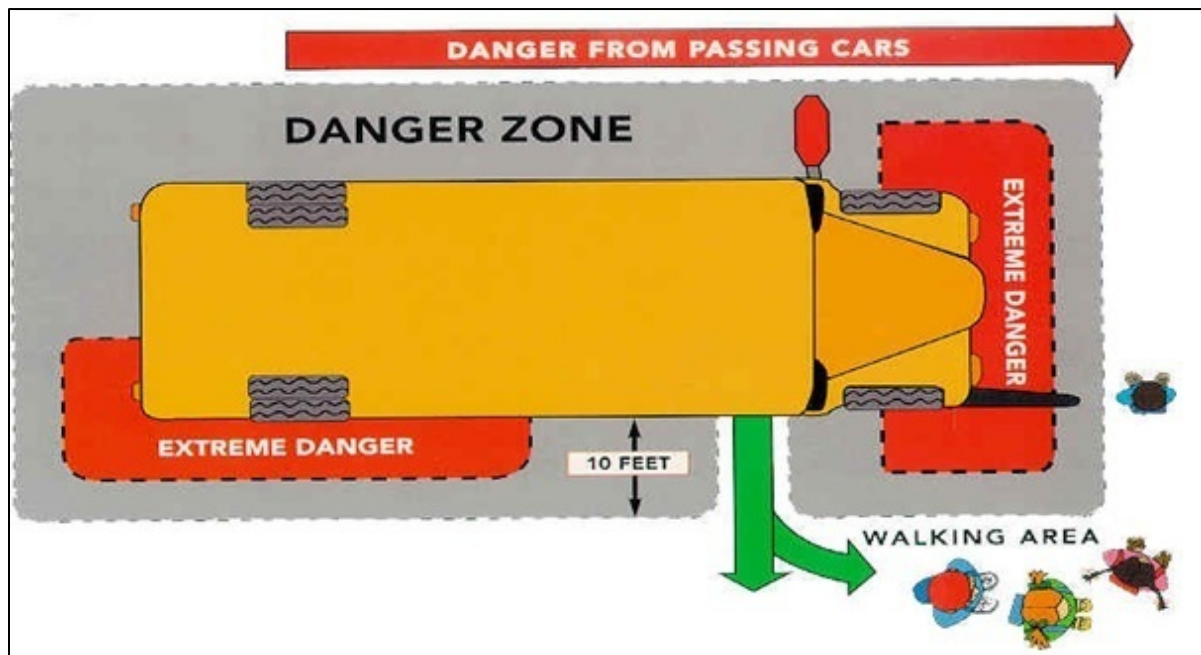
3.1 Selecting Appropriate Camera

When selecting cameras for a bus several elements must be taken into consideration.

- Child safety internal
- Child safety external

The image below shows an example of possible external danger zones for students. Primarily due to visibility of the students to the Bus Driver, and any on-coming vehicles.

As you can see from the image below there are numerous potential position on a bus where a camera could be located. That is where better understanding of camera focal points, features, and options will help you recommend the right camera for the right location.



For example: You can select 1 or 2 cameras for minimal coverage or up to 12 depending upon system and how you wish to maximize the views in and around your vehicle. Our next section will cover this in greater detail.

3.2 Camera Placement Internal and External

Why place a camera there? That is a great question. To sum it up, it has everything to do with the best camera for the application. In other words, where is it going to be placed, and what conditions will it be exposed to? Will it be viewing multiple lanes, exposed to changing temperatures, lots of vibrations, etc.?

Cameras can be mounted anywhere in the vehicle that provides a stable mount or and does not vibrate excessively. Additionally, use only outdoor cameras for exterior placement.

All cameras come standard with:

- High-resolution color images.

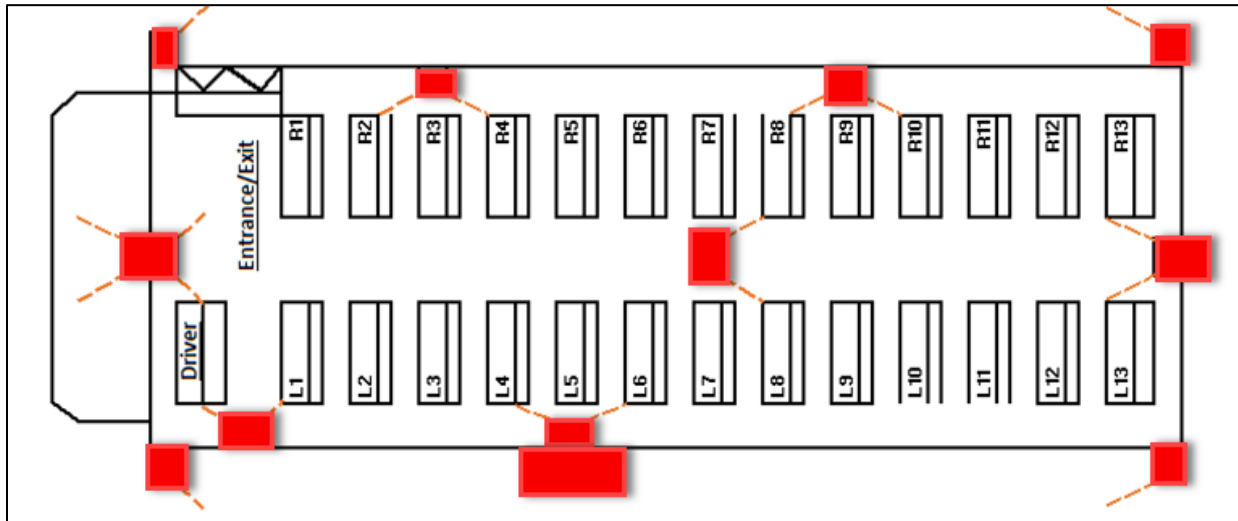
And most offer:

- Day or Night viewing capabilities
- Microphone for audio
- Intelligent IR technology
- Adjustable viewing angles.
- Built-in heaters
- Tamper resistance
- Waterproof
- And low profile

Camera Placement

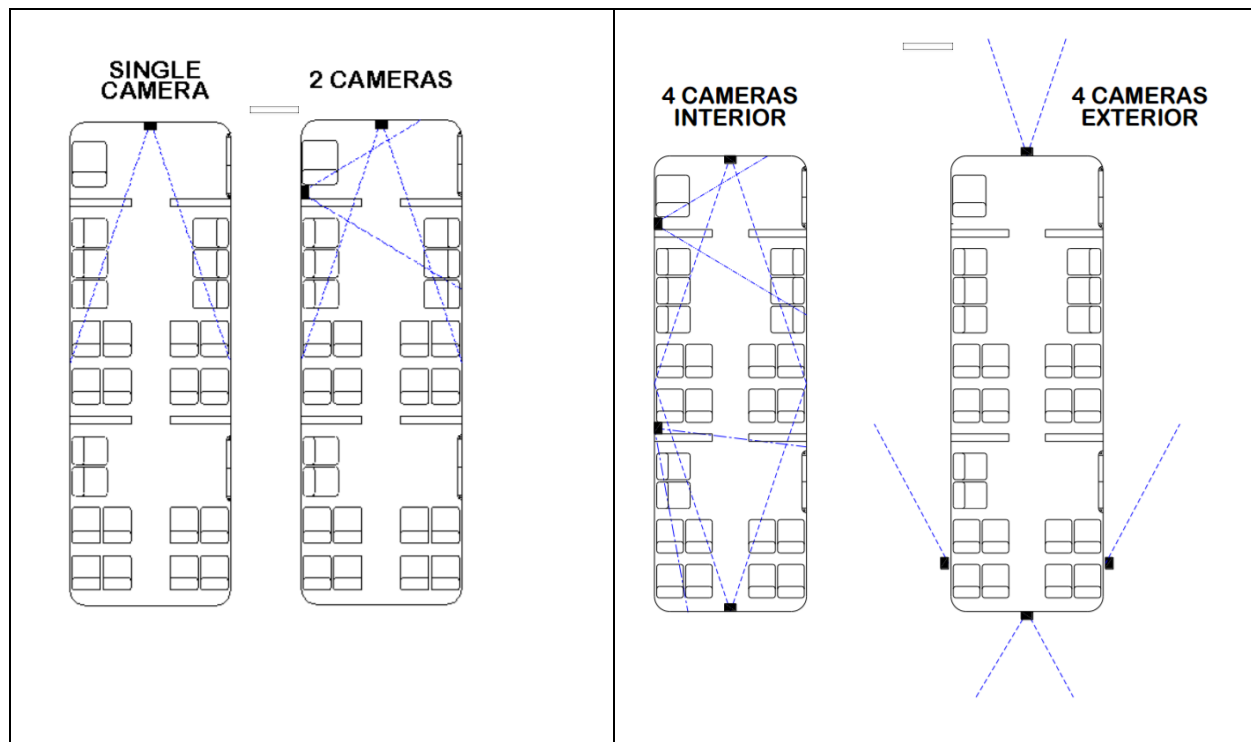
When it comes to camera placement, this next section will focus the distinct types of cameras offered and where to place them on a bus.

Multi-Camera Placement Options on a School Bus



For example: Next to the driver seat looking in. The options best suited for this are:

- 710502 - CAM,MB,HR650,2.4MM,DAY/NIGHT, AUDIO
- 710350 - CAM,MB,HR650,2.8MM,DAY/NIGHT, AUDIO



Camera Placement Practice

Using the camera options provided in section 3, please select options appropriate for the following positions within a school bus.

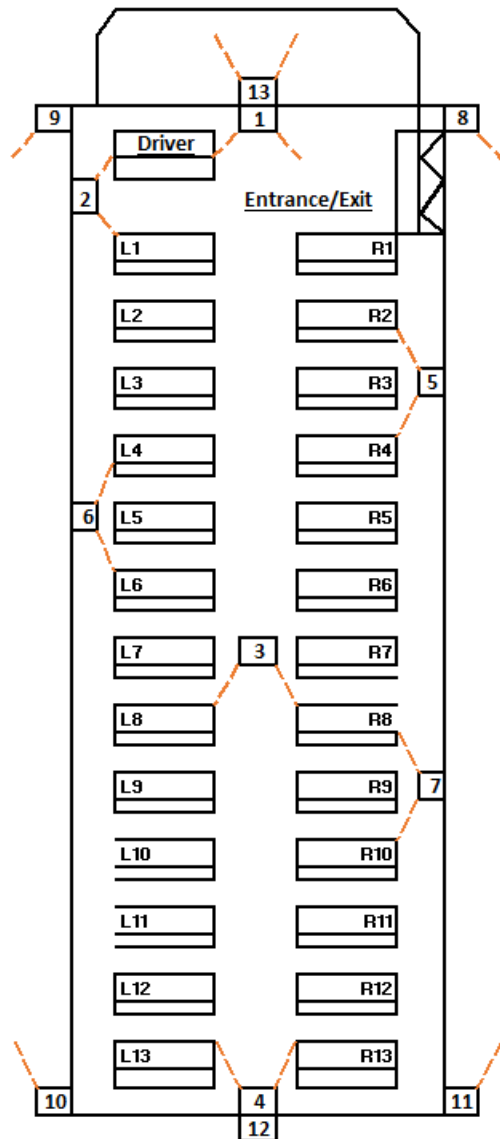
	Camera 1	
	Camera 2	
	Camera 3	
	Camera 4	
	Camera 5	
	Camera 6	
	Camera 7	
	Camera 8	
	Camera 9	
	Camera 10	
	Camera 11	
	Camera 12	
	Camera 13	
	Camera 14	
	Camera 15	
	Camera 16	

Optimal System Placement and Quote Graph

Optimal System Placements and Quotes																							
DVR	HardDrive	Camera#1	Camera#2	Camera#3	Camera#4	Camera#5	Camera#6	Camera#7	Camera#8	Camera #9	Camera #10	Camera #11	Camera #12	Camera #13	Event Marker	GPS	G-Force	Sensor Harness	J1939	STOPCAM	Wi-Fi Antenna	Cell Antenna	Wi-Fi + Cell
SD40	32GB	X			X																		
SD40	64GB	X	X		X									X	X								
DR40	500GB	X	X	X	X										X	X							
DR40	1TB	X	X		X							X			X	X		X					
HD420	500GB	X	X		X										X								
HD420	1TB		X			X	X	X							X	X	X	X					
HD420W	500GBSSD		X			X	X	X							X	X	X	X	X		X		
HD420W	1TBSSD	X	X	X											X	X				X	X		
HD800	1TB		X			X	X	X						X	X	X	X	X	X				
HD800	1TBSSD		X			X	X	X				X	X							X			
HD800	2TB		X			X	X	X		X	X	X	X		X	X	X	X					
HD800W	1TBSSD	X	X			X	X	X	X	X					X					X		X	
HD800W	1TB	X	X			X	X	X	X	X					X	X	X			X	X		
HD800W	1TBSSD	X	X	X	X						X	X	X		X	X	X	X	X				X
HD800W	2TB	X	X	X	X	X	X								X	X	X	X	X	X	X		

*Yellow Fields indicate exterior cameras and will need exterior cables.

Bus Camera Preferred Placement



Camera #	Type
Camera 1	710350, 710351, 710352
Camera 2	710502, 710350
Camera 3	710370, 710501
Camera 4	710350, 710351, 710352
Camera 5	710502
Camera 6	710502
Camera 7	710502
Camera 8	710560, 710561
Camera 9	710567, 710568
Camera 10	710560, 710561
Camera 11	710567, 710568
Camera 12	710311

Notes:

4 Software

This next section will be devoted to the different software options currently offered through REI. Beginning with the V.M.S. – Video Management Software, which is essentially a viewing platform for our hard-drives. Then wrapping up the software training with a brief introduction into our A.R.M.O.R. Asset Reporting and Management/Observation software.

4.1 V.M.S. – Video Management Software

- **V.M.S. – Video Management Software** - The HD Series Software (VMS) is used to view, save, and capture video from the HD Series DVR. This is a complementary/free download located at the following site:

To download the new HD5 VMS:

- **HD5 VMS Download:** [F:\San Diego Airport\622011 - SOFTWARE, HD5 VMS\REI VMS\(2.0.3.51\) English.exe](F:\San Diego Airport\622011 - SOFTWARE, HD5 VMS\REI VMS(2.0.3.51) English.exe)

To begin download:

1. Go to Customer Support Site: <http://www.radioeng.info/index.html>
2. Locate and double click the icon, “**Setup REI VMS.exe**”.
 - i. If received on a CD-ROM, this program will be in the root directory of the CD-ROM. Your setup may begin automatically upon insertion of the CD-ROM.
 - ii. If received via a download from the web, this program will be in the directory your web browser or FTP client downloaded it to. Please refer to the documentation of your download program to find it. If the User Account Control screen is shown, click “**Yes**”.
3. If the **User Account Control** screen is shown, click “**Yes**”.
4. When the “**Welcome Screen**” shows, follow the instructions on the screen.
5. Choose where to install REI VMS. It is recommended you leave the default location and click “**Next**”.
6. Choose what you will call the application in the Start Menu. It is recommended you **leave the default name** and click “**Next**”.
7. Select whether you wish to create a desktop shortcut for REI VMS. It is recommended you **leave this checked** and click “**Next**”.
8. Click “**Install**”
9. Wait for installation to complete. On the final screen click “**Finish**”.

Your REI installation will have installed the following programs:

- **REI VMS** - This application is used to search, view, and/or backup video stored on the hard-drive module, stored on a DVR, or for viewing previously downloaded or captured video stored on the host PC. It also provides for the simultaneous viewing (monitoring) of multiple channels of live video from multiple networks connected to mobile DVRs.

- **REI Image Viewer** - This application is used display and share snapshots of video taken by REI VMS. This provides a convenient way to edit, save, and share snapshots taken from REI-compatible programs.
- **REI Express** - This is a standalone application used for sharing and distribution of .BWV video files created by REI VMS. You may include this with any .BWV files in order for people who do not have the REI VMS installed to view video.

4.2 A.R.M.O.R.



- **A.R.M.O.R.** - Asset Reporting & Management/Observation & Recording software is not a free download. The A.R.M.O.R. package is customizable to the needs of the customer. For example, we can manage the following areas:
 - User Credentials
 - DVRs
 - Vehicles
 - Vehicle Groups
 - Event notification
 - Driver schedules
 - And much more.

We will discuss A.R.M.O.R. in greater detail in our specific section regarding A.R.M.O.R.

5 Accessories

This last section will briefly show the many different accessories currently available for purchase. When it comes to systems there are some important additional accessories that may need to be added to the system. Those accessories include:

Wi-Fi Antenna / Cable 	Inertia Sensor Accelerometer 
GPS Receiver 	Panic Button / Event Marker 
Security Caps 	Cellular Antennas 
Broadband Router 	Camera Observation System 
SD Cards 	SD Card Reader 